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RESEARCH

Evaluating the effect of cryotherapy and low-level laser therapy on postoperative pain and quality of life in patients with symptomatic apical periodontitis- a randomized controlled clinical study

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Abstract

The aim of this randomized controlled clinical study is to compare the effect of intracanal cryotherapy and intraoral Low-Level LASER Therapy (LLLT) applications on postoperative pain and quality of life in patients following endodontic treatment for symptomatic apical periodontitis. After ethical clearance and registering the trial at clinical trial registry of India, a randomized, parallel-controlled clinical study with 2 test arms and 1 control arm was conducted. 90 subjects diagnosed with acute apical periodontitis in molar teeth meeting the inclusion and exclusion criteria, were enrolled in the current trial. After obtaining written informed consent, first researcher randomly allocated the subjects into 3 arms: intracanal cryotherapy, intraoral Low-Level LASER Therapy (LLLT) and control. Preoperative pain scores were marked on the Heft- Parker visual analog scale (HP-VAS) by the second researcher. Preoperative quality of life was marked by the subjects on the Oral Health Impact Profile-17 (OHIP-17) questionnaire. Root canal treatment was performed for all the subjects enrolled in the study and the interventions were performed accordingly, by a single operator. The patients were instructed to record their postoperative pain levels and analgesics intake at 24 h, 48 h, the third day, fifth day, and seventh day. On the seventh day postoperative quality of life questionnaire was recorded by the subjects. The data were tabulated in MS Excel and statistically analyzed using the EPI-INFO software. After 24 h, postoperative pain reduced significantly in the cryotherapy arm when compared with the LLLT arm. However, no statistically significant differences in postoperative pain were found between the two test arms at 48 h, the third day, fifth day, and seventh day. Subjects in the control arm were relieved of postoperative pain after the fifth day. Subjects in the cryotherapy arm required fewer analgesics, with intake limited to the first 24 h postoperatively. Conversely, participants in the LLLT arm needed analgesics for the initial 48 h postoperatively. In contrast, subjects in the control arm had the longest duration of analgesic use, up to the fifth day. The quality of life for all the subjects was found to improve postoperatively after seven days, irrespective of additional interventions. Application of cryotherapy and Low-Level LASER Therapy was found to be effective at 24 h and 48 h respectively, in the management of postoperative endodontic pain. The findings highlight the potential benefits of cryotherapy and LLLT compared to standard postoperative care alone. Henceforth, cryotherapy and Low-Level LASER Therapy could be utilized, as they are simple and innocuous modalities to manage post-endodontic pain.

Keywords Cryotherapy · Low level LASER Therapy · Post Endodontic Pain · Post operative Pain · Quality of Life · Symptomatic Apical Periodontitis

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Introduction

Endodontic treatment alleviates pain in the long-term nevertheless postoperative pain immediately after following root canal treatment is a common adverse effect, resulting in significant patient dissatisfaction [1]. A survey by the American Association of Endodontists on the public's perception of endodontic treatment was found to be negative, as the treatment was associated with pain before, during, and after treatment [2]. A study reported that one in ten adult patients who underwent root canal treatment experienced postoperative pain or discomfort [3]. DiRenzo and colleagues reported in their review that the incidence of postoperative pain after non-surgical endodontic treatment could exceed 50% [4]. This inevitably affects quality of life [5].

Endodontic science is dedicated to improving patients' quality of life by providing comfort and safety through its state of the art techniques. Effective prevention and management of such postoperative pain are essential to enhance patient outcomes. Comprehensive knowledge of pain after root canal treatment enables clinicians to predict and effectively adopt strategies to manage such challenging situations [1, 6]. Various pharmacological and non-pharmacological approaches have been employed to manage postoperative endodontic pain. Owing to the adverse effects of pharmacological agents and the limitations of their use in certain patients, relatively safer and more efficacious alternative treatment modalities such as cryotherapy and Low-Level LASER irradiation have been considered for the management of postoperative endodontic pain [7, 8].

The term cryotherapy is derived from the Greek word cryos, meaning "very cold"or "icy cold", referring to the use of low temperatures for therapeutic purposes [9]. Since the early 1960 s, cryotherapy has been used clinically for pain management [10]. In the field of dentistry, cryotherapy has been proven to be effective in reducing pain and swelling following intraoral excisional surgical procedures, periodontal surgery, tooth extractions, implant placement, and in the management of arthritis associated with temporomandibular joint disorders [11, 12]. Low-Level LASER Therapy (LLLT)is one of the approaches of photobiomodulation therapy that has been employed for the modulation of inflammation, regeneration and pain control at the molecular level [13]. Since 1970, it has been utilised in dentistry to treat conditions such as recurrent herpes, aphthous stomatitis and to alleviate postoperative discomfort [14].

Quality of life (QoL) is defined by the World Health Organization as 'individuals' perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards, and concerns'' [14]. The efficacy of root canal treatment is typically determined by clinical outcomes and tooth retention; however, this does not take into account the patient's perspective [15, 16]. The present trial evaluated.

the impact of cryotherapy or Low-Level LASER application during root canal treatment on the quality of life of patients.

A thorough literature search to the best of our knowledge revealed that there were no studies in the literature that compared LLLT and cryotherapy applications for postoperative pain reduction following initial endodontic treatment. Therefore, the present clinical trial was designed with the aim of evaluating the effect of intracanal cryotherapy and intraoral Low-Level LASER Therapy (LLLT) applications on postoperative pain and quality of life following endodontic treatment in patients with symptomatic apical periodontitis. The hypothesis of this study was that there would be a difference in the effect of intracanal cryotherapy and intraoral Low-Level LASER Therapy (LLLT) applications on postoperative pain and quality of life in patients following endodontic treatment for symptomatic apical periodontitis.

Methodology

The study was designed as a randomized controlled clinical trial with a parallel design. After being independently reviewed by the Institutional Ethics Committee, the certificate of approval to conduct the research was issued under code no. IEC/SVDCH/1902. The trial was registered in the Clinical Trial Registry- India with the registration Id (CTRI/2021/02/030939||http://ctri.nic.in/) on 02.02.2021. The trial was conducted in the Department of Conservative Dentistry and Endodontics, Sri Venkateswara Dental College and Hospital, following ethical principles.

The sample size was calculated using Open Epi software (version 3), with data derived from previously conducted trials [17–19]. The nature and purpose of the trial were first explained to the subjects in English and the regional language. Ninety subjects meeting the inclusion and exclusion criteria were enrolled for the trial. Subjects were then enrolled with the understanding that written consent was obtained from them individually.

Subjects aged between 18 and 50 years, with symptomatic apical periodontitis of vital maxillary or mandibular molar teeth, requiring primary endodontic treatment and with a pain level on the Heft Parker Visual Analogue Scale (HP-VAS) score > 114 were included in the study. Subjects with systemic diseases, allergic reactions, pregnancy, subjects who have taken analgesics in the last three days, subjects having pain in a previously root canal initiated tooth, non-vital teeth, teeth with sinus tract and periapical radiolucency, subjects with severe periodontal disease, swelling, periodontal pocket depth > 3 mm, subjects with

parafunctional habits, occlusal discrepancies, smokers, alcoholics, tobacco chewers and subjects under medications for anxiety, psychosis, sedatives, hypnotics or antidepressants were excluded from the study.

The study consisted of two test arms and one control arm, with thirty subjects enrolled in each arm. Subjects in the Test arm I received intracanal cryotherapy before obturation, while subjects in Test arm II received Low-Level LASER Therapy (LLLT) immediately following obturation. Subjects in the control arm were not subjected to LASER or cryotherapy applications.

Subjects were randomly allocated into three arms using a random-allocation software (Research randomizer, Version 4.0, randomizer.org). A computer-generated list of random numbers was used for subject allocation to each group with a randomization ratio of 1:1:1. Randomization was performed by the first researcher. The group allocation of each subject was individually placed in opaque, sealed, and numbered envelopes and given to the respective subjects. The allocation sequence was concealed from the second researcher, who recorded the pain scores of the subjects before and after treatment. After cleaning and shaping of the root canals, the first operator opened the envelope to check for the group allocation and proceeded accordingly. The pain scores were recorded by the patient using the HP-VAS pain scale under the supervision of the second researcher, who was not aware of the group to which they belonged.

Preoperative data for each subject was recorded in the predesigned patient's chart by the second researcher. The second researcher tested the subjects teeth using the cold test (Endo-frost, Coltène, Whaledent, Langenau, Germany) and an electric pulp tester (EPT) (Sybron Endo Vitality Scanner Pulp Tester, CA, USA). A single operator performed root canal treatment under 2% Lignocaine with 1:200,000 epinephrine local anesthesia (Indoco remedies Ltd.,Palghat, India). The treatment was initiated after pulpal anaesthesia was ensured using the EPT, eliciting no response to maximal output on two consecutive tests at least two minutes apart.

Under rubber dam (GDC dental dam-medium, India) isolation, a straight-line access cavity was prepared using Endo-access bur size 2 (Dentsply Maillefer, Ballaigues, Switzerland) with an air turbine hand piece (NSK, Tochigi, Japan) under copious water cooling. The working length was determined using an apex locator (J Morita root ZX mini, Europe GVBH, Frankfurt, Germany) and confirmed a with digital radiograph after ensuring canal patency with a stainless steel 10 size K file (Mani, Utsunomiya, Tochigi, Japan). The Crown down technique was used to prepare the canals up to size 20, 6% or 25, 6% using an endomotor (speed: 250 rpm, torque: 150 N) (X-Smart; Dentsply) and rotary files (Neoendo flex files, Orikam health care pvt. Ltd, India). Irrigation was performed using a 30-gauge side-vented needle (Neoendo flex files, Orikam health care pvt. Ltd, India) with 10 ml of 3% sodium hypochlorite (Vensons, Bangalore, India) for 60 s per canal. The sealed randomization envelope was then opened by the operator to check for the group allocation of the respective subject.

For the subjects belonging to Test arm I, final flushing of the root canals with 20 mL cold saline solution (2.5 °C) (Fresenius Kabi India Pvt. Ltd, Goa, India) for 5 min (~2 min/canal) was done followed by drying with absorbent paper points (DiaDent, Canada). Syringes loaded with saline were kept in a calibrated refrigerator at 2.5 °C until use [12]. Obturation was done by cold lateral compaction using gutta percha cones (Dentsply Sirona Tulsa Dental, York, PA, USA) and AH plus sealer (Dentsply Maillefer, Ballaigues, Switzerland). The access cavity was sealed incrementally using composite resin (Tetric N- Collection system kit, Ivoclar Vivadent, Liechtenstein).

For the subjects belonging to Test arm II (LLLT group), intervention was done after obturation and access sealing. LLLT was performed using a 970 ± 15 nm diode LASER (SIROLASER Xtend; Sirona Dental Systems GmbH, Bensheim, Germany). The fibre tip was placed at a distance of approximately 10 mm from the mucosa overlying the root apex and activated at 0.5 W and 10 Hz (power density 286 W/cm2). The buccal and the lingual or palatal mucosa overlying the apices of the mesial, distal, buccal and palatal roots of the tooth were irradiated for 30 s per site. LASER irradiation was done for a total of 120 s per tooth in the mandible and 90 s per tooth in the maxilla. For the subjects in the control arm conventional non-surgical endodontic treatment was performed. Neither intervention with cold saline nor Low-Level LASER application was done. Final radiographs were taken. All the subjects were instructed to use 400 mg Ibuprofen if symptomatic and record their intake. All the subjects belonging to the test arms were subjected to the intervention only once during the endodontic treatment. The endodontic treatment was completed in a single-visit for all the subjects in this study.

Pain assessment

Pain Assessment was done using Heft-Parker Visual Analogue Scale (Fig. 1) which is a 170-mm line with descriptive terms. Each subject was educated and instructed to place a mark on the scale where it best described their pain level. Each subject completed a baseline HP VAS score to establish their preoperative pain level. A separate form was provided to the subject after treatment to record their postoperative pain and analgesic intake on the first, third, fifth, and seventh days. Any unscheduled emergency inter-appointment pain score was recorded by the second researcher.



Fig. 1 showing the Heft Parker Visual Analogue Scale (HP VAS Scale)

Quality of life assessmentThe modified Oral Health Impact Profile instrument (OHIP-17) questionnaire was provided before root canal treatment and a week after completion of the treatment. Participants were asked to rate the occurrence of an event as described by the OHIP-17 on a 5-point Likert scale. The scoring was as follows 'strongly disagree' (score 5), 'disagree' (score 4), 'neutral' (score 3), 'agree' (score 2) and 'strongly agree' (score 1).

All the patient demographic details, pain scores and quality of life assessment scores were recorded in the patient data recording sheet. The obtained data were then tabulated in Microsoft excel by the second researcher and analysed using EPI-INFO (version 7.2.2.6, CDC license: Atlanta, Georgia US: public domain) software. The statistical tests employed were as follow: For inter-group comparison of pain scores, the Kruskal Wallis test was used while for intra-group pain score comparison, Friedman's analysis was used. For Quality-of-life assessment, Wilcoxin signed-rank and Mann-Whitney U tests were used. For analgesics intake analysis, Kruskall Wallis and Mann-Whitney U tests were used.

Results

Among the ninety subjects enrolled for the trial, two subjects did not report for the postoperative evaluation after 24 h, one subject did not report from the third day and two subjects dropped out from the fifth day. Among the ninety subjects enrolled, eighty five subjects [38 (44.7%) male and 47 (55.3%) female] completed the trial. The attrition rate was found to be 4.5%. This attrition rate is below the 10% threshold set during the sample size calculation. The subjects enrolled for the current trial belonged to the age group 18–48 years, with a mean age of 36.15 ± 10.5. Table 1 summarizes the demographic characteristics of all the three arms. Among the three arms, there were no significant differences observed in the preoperative pain scores of the subjects enrolled (p-value = 0.219) as shown in Table 2.

Over time within the three arms, there was a statistically significant difference in the pain scores as seen in Table 3. Subjects in the cryotherapy and LASER arms experienced a reduction in postoperative pain scores from 24 h onwards. However, subjects belonging to the control arm reported pain till the fifth day. The results demonstrated that there was a statistically significant difference in the pain scores among the three arms at all postoperative time intervals except on the seventh day, as shown in Table 4.

Table 4 summarizes the Post hoc comparison of the pain scores at different time intervals in each arm, while Table 5 shows the Post hoc comparison of pain scores among the three arms at different time intervals. At 24 h, cryotherapy intervention was found to perform better than LASER in postoperative pain reduction. After 48 h onwards, LASER and cryotherapy interventions performed similarly and were better than the control arm. On the seventh day, the subjects belonging to all three arms reported no postoperative pain, while all other comparisons showed that there were no statistically significant differences.

Analgesics intake

Table 6 depicts the mean analgesic intake required by the subjects in each arm at different time intervals (24 h, 48 h,

Table 1 Demographic characteristics of the subjects enrolled in the trial

Variables	LASER	Cryotherapy	Control
N	28	29	28
Mean (S.D)	32.96 +/- 10.27	37.69 +/- 10.06	37.82+/-10.90)
Male	15	13	10
Female	13	16	18
Total	28	29	28

Table 2 Pre-operative pain score comparison among three arms

Arm	*N*	Mean Rank	Chi-Square	df	*p*-value
Cryotherapy	29	48.66	3.037	2	0.219
Laser	28	38.68			
Control	28	41.46			

Table 3 Intra group and inter group comparison of pre and post-operative pain score at different time intervals (24 h, 48 h, 3rd day, 5th day & 7th day)

Post-operative time	Cryotherapy mean rank	LASER mean rank	Control mean rank	*p*-value
24 h	23.72	35.00	70.96	0.001*
48 h	27.28	32.61	69.68	0.001*
3rd day	31.00	32.43	66.00	0.001*
5th day	39.00	39.00	51.14	0.001*
7th day	42.00	43.52	43.52	0.592
p-value	0.001*	0.001*	0.001*	

*Denotes significance at 1% level

Table 4 Post hoc comparison of the pain scores at different time intervals (24 h, 48 h, 3rd day, 5th day & 7th day) within the three arms

Pain score comparison	Cryotherapy	Laser	Control
Pre-op& 24 h	0.001*	0.001*	0.001*
Pre-op& 48 h	0.001*	0.001*	0.001*
Pre-op& 3rd day	0.001*	0.001*	0.001*
Pre-op& 5th day	0.001*	0.001*	0.001*
Pre-op& 7th day	0.001*	0.001*	0.001*
24 h & 48 h	0.001*	0.001*	0.001*
24 h & 3rd day	0.001*	0.001*	0.001*
24 h & 5th day	0.001*	0.001*	0.001*
24 h & 7th day	0.001*	0.001*	0.001*
48 h & 3rd day	0.001*	0.001*	0.001*
48 h & 5th day	0.001*	0.001*	0.001*
48 h & 7th day	0.001*	0.001*	0.001*
3rd day & 5th day	0.001*	0.001*	0.001*
3rd day & 7th day	0.001*	0.001*	0.001*
5th day & 7th day	0.001*	0.001*	0.001*

Denotes significance at 1% level

Table 5 Post hoc comparison of pain scores among the three arms after 48 h

Arms		*p*-value
24th Day	Laser * Cryotherapy	0.003*
	Cryotherapy * Control	0.001*
	Laser * Control	0.001*
48th Day	Laser * Cryotherapy	0.079
	Cryotherapy * Control	0.001*
	Laser * Control	0.001*
3rd Day	Laser * Cryotherapy	0.309
	Cryotherapy * Control	0.001*
	Laser * Control	0.001*
5th Day	Laser * Cryotherapy	1.000
	Cryotherapy * Control	0.002*
	Laser * Control	0.002*

Table 6 Inter group comparison of analgesic intake among three arms at different time intervals (24 h, 48 h, 3rd day, 5th day & 7th day)

Group		Mean Rank	*p*-value
24 hrs	LASER	36.25	0.001*
	Cryotherapy	23.28	
	Control	70.18	
48 hrs	LASER	30.61	0.001*
	Cryotherapy	27.50	
	Control	71.45	
3rd day	LASER	35.00	0.001*
	Cryotherapy	35.00	
	Control	59.29	
5th day	LASER	41.00	0.001*
	Cryotherapy	41.00	
	Control	47.07	
7th day	LASER	43.00	1.000
	Cryotherapy	43.00	
	Control	43.00	

third day and fifth day). On comparing the analgesic intake among the three arms at these different time intervals (24 h, 48 h, third day and fifth day), differences were found to be statistically significant, as shown in Table 6.

Quality of life (QOL) assessment

Subjects in all the three arms showed improvement in their quality of life following endodontic treatment during the one-week follow-up period. From the clinical data, it can be inferred that the cryotherapy arm had the highest score, followed by the LASER and control arms. Hence, it can be deduced that the subjects quality of life belonging to the cryotherapy arm was better when compared to the subjects in the LASER and control arms.

Discussion

Despite pain being the major reason for seeking treatment, the fear of experiencing it post-treatment still remains a major cause for the denial of root canal therapy [20]. Suspending the need for endodontic treatment is a clear indication that it has an impact on the quality of life. The inflammatory mediators produced by means of mechanical, chemical, and microbial injuries to the pulp or periradicular tissues during endodontic treatment activate the sensitive nociceptors and lead to central and peripheral hyperalgesia mechanisms thus resulting in the perception of post-treatment pain [21, 22].

Novel methods have been proposed in an endeavour to offer greater comfort to patients by reducing the postoperative pain [23]. These include preoperative relaxation approaches with explanations for patients, the use of different kinematics during root canal treatment, intracanal LASER irradiation, cryotherapy application and Low-Level LASER Therapy (LLLT) [24–27]. These methods have been shown to decrease the intake of analgesics, thereby minimising the side effects. Additionally, they have also been found to improve the patients' quality of life [28, 29].

Cryotherapy functions by extracting heat from the tissue rather than merely cooling the target tissue [30]. The three basic physiological responses of tissues following the application of cold temperatures are a reduction in local blood flow, inhibition of neural receptors in the skin and subcutaneous tissues, and a decrease in metabolic activity [31]. In endodontics, cryotherapy has been reported to be used during root canal treatment and following periradicular surgeries to minimise inflammation, thereby reducing the postoperative pain [32]. Al-Nahlawi et al.(2016) and Keskin et al.(2017), have studied the clinical use of cold saline following single-visit root canal treatments and found it to be effective in reducing postoperative pain [25, 33].

Gundogdu et al.(2018) evaluated the efficacy of intracanal, intraoral, and extraoral cryotherapy on postoperative pain and reported that all cryotherapy groups exhibited significantly lower levels of pain postoperatively, irrespective of the mode of application [19]. Several pieces of evidence support cryotherapy as a simple, cost-effective, and non-toxic technique frequently used during various procedures for postoperative care and pain management [34, 35].

LASERs can be classified into two types based on their power as high and low. High power/surgical LASERs are usually employed to cut, vaporize, or resect tissues. However, low power LASERs produce photobiological effects within the cellular matrix and the mitochondria. This photobiomodulation/biostimulation accelerates healing, increases blood circulation, decreases oedema, and minimises pain. It has been reported that LASER wavelength is the main factor determining penetration depth. In dentistry, a range of 600–1000 nm in the near-infrared spectrum with penetration depths of upto 5 mm is frequently employed [36]. Therefore, it is found to affect the periapical tissues [37, 38]. LLLT application duration is a crucial factor. In dentistry, LLLT application times are within the 30–60 s range per treatment site [39]. Studies have shown that buccal and lingual irradiation are more effective in reducing postoperative endodontic pain than only buccal irradiation [40]. Therefore, for this trial, a 970+/−15 nm diode LASER was used with a fibre tip placed at a distance of approximately 10 mm from the alveolar mucosa overlying the root apex and activated at 0.5 W and 10 Hz. The buccal and the lingual mucosa overlying the apices of the mesial, distal and palatal roots of teeth were irradiated for 30 s per site [17, 27].

Cryotherapy application was more effective in reducing the postoperative pain than LLLT application at 24 h. However, both interventions performed similarly in reducing the postoperative pain after 48 h onwards. Postoperative pain reduced gradually decreased over 7 days in all three arms, irrespective of the intervention. Clinically, subjects belonging to the cryotherapy arm required the least number of analgesics, only during the first 24 h. Subjects exposed to LLLT required analgesics for up to 48 h postoperatively, whereas the subjects in the control arm required the highest number of analgesics postoperatively for up to the fifth day.

It is important to understand the burden of disease and the benefits of care provided in the daily lives of the patients. Quality of life instruments measure the population‘s perception of the impact of oral disorders on their well-being and thus are a valuable tool [14]. In 1994, the Oral Health Impact Profile (OHIP) was developed to measure the social impacts of oral disorders. This model was designed to identify seven conceptual dimensions of impact [14]. A shorter version of the Oral Health Impact Profile was designed by narrowing the original instrument’s 49 items to 17 items, relating to endodontic disease by Dugas et al. The modified form of the OHIP quality-of-life instrument has been proven to have high validity and reliability [41]. Studies have been conducted to identify key factors associated with the impact of endodontic disease on overall health and oral health–related quality of life (OHRQoL) [20, 41, 42]. The evaluation and comparison of preoperative QoL and postoperative QoL (POQoL) represent more standardized outcome assessment methods [14–16]. A regional study concluded that conventional endodontic treatment improved the quality of life score by 60% and revealed that the patients recognized the negative impact of endodontic disease on their quality of life [16].

The quality of life of all the subjects improved following root canal treatment after a week, irrespective of the interventions. This was a limitation of this trial, as postoperative QoL was assessed only on the seventh day. However, clinically the quality of life of the subjects belonging to the cryotherapy arm was better than that of the LASER arm and the LASER arm was better than control arm respectively. Clinically the QoL of the subjects in the cryotherapy and LLLT arms improved at an earlier time point when compared to the subjects in the control arm, as the control arm subjects were relieved of postoperative pain only after fifth day.

Conclusion

Within the limitations of the present study, it can be concluded that both cryotherapy and Low-Level LASER Therapy applications were found to be superior to conventional endodontic treatment without any intervention in effectively managing postoperative endodontic pain. 24-four-hour cryotherapy was found to be superior to Low-Level Laser Therapy application in terms of reduced postoperative pain and improved QoL. Furthermore, well-designed trials are required to focus on better methodologies for the determination of optimum LASER parameters for enhance pain control and management.

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Author contributions The study was designed by Dr. KungumaSupraja K. Dr. KungumaSupraja K, the principal investigator, acquired data along with Dr. Diana D and Dr. Duraivel D. The analysis & interpretation of data was done by Dr. KungumaSupraja K, Dr. Srinivasan MR and Dr. Poorni S. Drafting of the article was done by Dr. KungumaSupraja K and Dr. Diana D. Revision of the article was done by Dr. Poorni S, Dr. Duraivel D and Dr. Srinivasan MR.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Human ethics and consent to participate The study proposal was reviewed by the Institutional Ethics Committee, and the certificate of approval to conduct research was issued under code no. IEC/SVDCH/1902.

Competing interests The authors declare no competing interests.

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